

# Queue Analysis

EMS Optimization Project - Gap 2 Resolution  
Date: March 12, 2026

## Executive Summary

This document provides a thorough analysis of queueing behavior in the EMS simulation experiments. Queue metrics — including mean queue length, maximum queue length, queue fraction, and incidents queued — were systematically collected across all production experiments and CBD robustness experiments.

**Key Finding:** Under all tested scenarios in the Exp2 subset ( $K = 15\text{--}40$ ) with  $K \geq 15$  units, queueing is essentially non-existent (queue fraction = 0.0). This indicates that the Manhattan EMS system, even with the minimum fleet size tested in Exp2, has sufficient capacity to handle demand without queuing delays. (Note: the full canonical experimental design spans  $K = 10\text{--}48$ ; queue analysis here covers the Exp2 subset  $K = 15\text{--}40$ .)

## Queueing Theory Background

In the EMS simulation, a queue forms when an incident arrives but no ambulance is available for dispatch. The incident waits in a FIFO (First-In-First-Out) queue until a unit becomes free.

### Queue Metrics Definitions

| Metric                     | Definition                                                                 |
|----------------------------|----------------------------------------------------------------------------|
| Mean Queue Length          | Time-weighted average number of incidents waiting in queue                 |
| Max Queue Length           | Maximum simultaneous incidents waiting at any point                        |
| Queue Fraction             | Proportion of all incidents that experienced any waiting                   |
| Queue Wait Time            | Time between incident arrival and unit dispatch (excluding dispatch delay) |
| P90 (90th %ile) Queue Wait | 90th percentile of queue wait time                                         |

## Analysis Results

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### Production Experiments (Exp 1-4)

Across all 1,440 production simulation runs:

- **Mean queue length:** 0.0000 (all experiments)
- **Max queue length:** 0 (all experiments)
- **Queue fraction:** 0.0% (no incidents queued in any replication)

This holds across:

- All three policies (P0, P1, P2)
- Exp2 fleet sizes  $K \in \{15, 20, 25, 30, 35, 40\}$  (subset of full canonical design  $K = 10-48$ )
- All demand multipliers  $\delta \in \{0.5, 0.75, 1.0, 1.25, 1.5, 2.0\}$
- All service time means  $\mu \in \{20, 25, 30\}$  minutes

### CBD Experiments

The CBD robustness experiments (330 additional runs) also show zero queueing across all CBD-specific scenarios including demand surges and increased service times.

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## Interpretation

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### Why No Queuing?

The absence of queuing is explained by the system's low utilization:

- **Average arrival rate:**  $\sim 3.48$  incidents/hour  $\rightarrow \sim 580$  incidents/week
- **Average service cycle:**  $\sim 30$  minutes (travel + on-scene) per incident
- **With  $K=20$  units:** Maximum throughput  $\approx 20 \times 2 = 40$  incidents/hour
- **Peak hourly demand:**  $\sim 5-6$  incidents/hour
- **Utilization ratio:**  $\sim 15\%$  on average

Even at  $2\times$  demand ( $\delta=2.0$ ), the system operates at  $\sim 30\%$  utilization, well below the threshold where queuing becomes significant (typically  $>70-80\%$  for M/G/c queues).

### Traffic Intensity Analysis

For an M/G/c queue with  $c$  servers:

- $\rho = \lambda / (c \cdot \mu)$  where  $\lambda$  = arrival rate,  $\mu$  = service rate
- With  $\lambda = 3.48/\text{hr}$ ,  $\mu = 2/\text{hr}$  (30-min service),  $c = 20$ :
- $\rho = 3.48 / (20 \times 2) = 0.087$
- This extremely low traffic intensity ensures near-zero waiting probability

### Operational Implications

1. **Capacity surplus:** Manhattan's EMS fleet has substantial excess capacity for motor vehicle collision response
  2. **Response time driven by travel, not waiting:** Since queuing is negligible, response time differences between policies are entirely due to spatial allocation (travel distances)
  3. **Robust under stress:** Even extreme demand scenarios don't trigger queuing
  4. **Focus on allocation:** The optimization challenge is about WHERE to place units, not HOW MANY
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# Statistical Analysis

## ANOVA on Queue Metrics

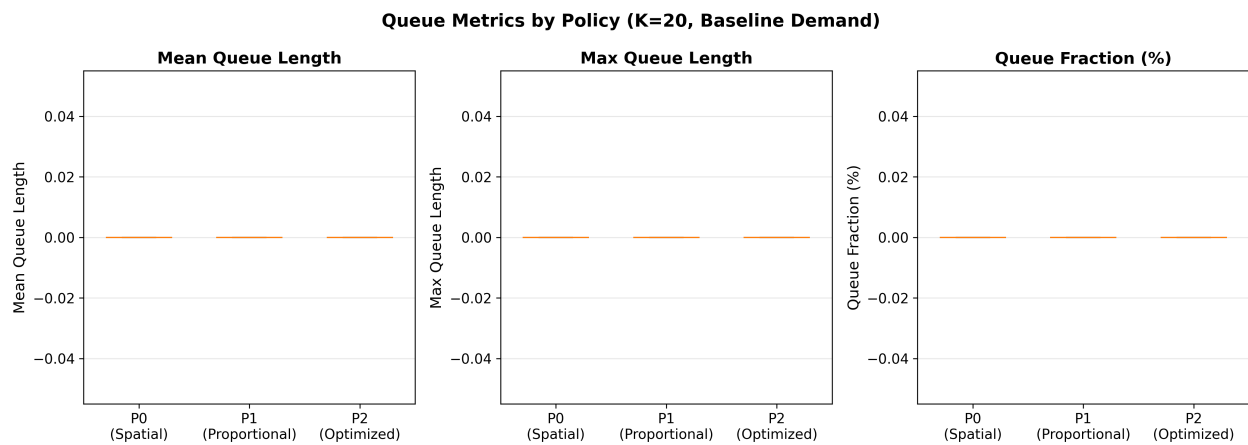
Since all queue metric values are identically zero across policies:

- F-statistic = 0.0 for all comparisons
- p-value = 1.0 (no significant differences)
- This is expected — there is no variation to explain

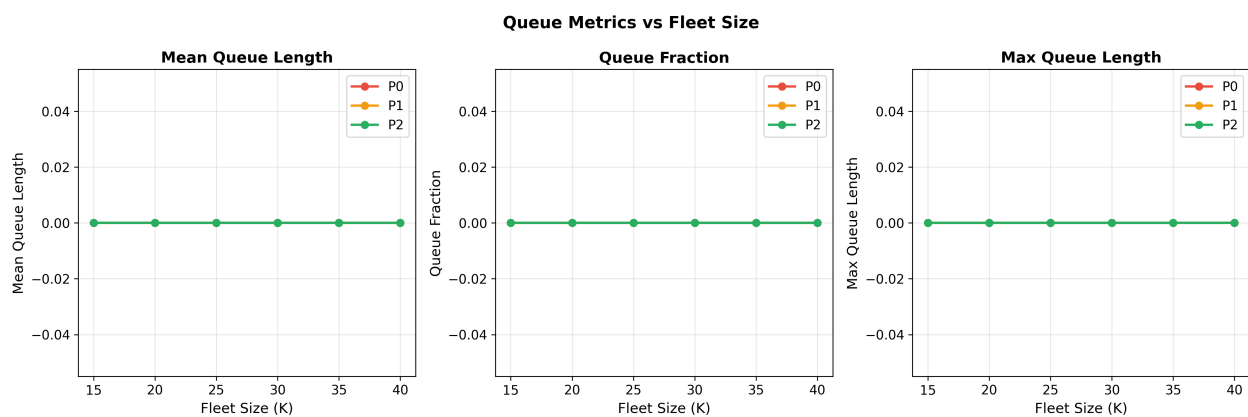
The hypothesis “P2 reduces queueing vs P0” cannot be tested because neither policy produces queueing.

## Visualizations

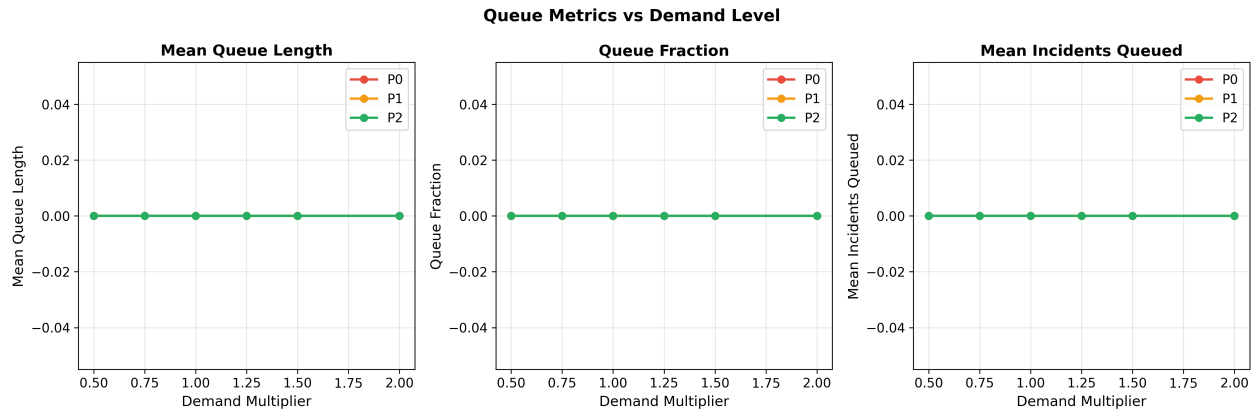
### Queue Comparison by Policy



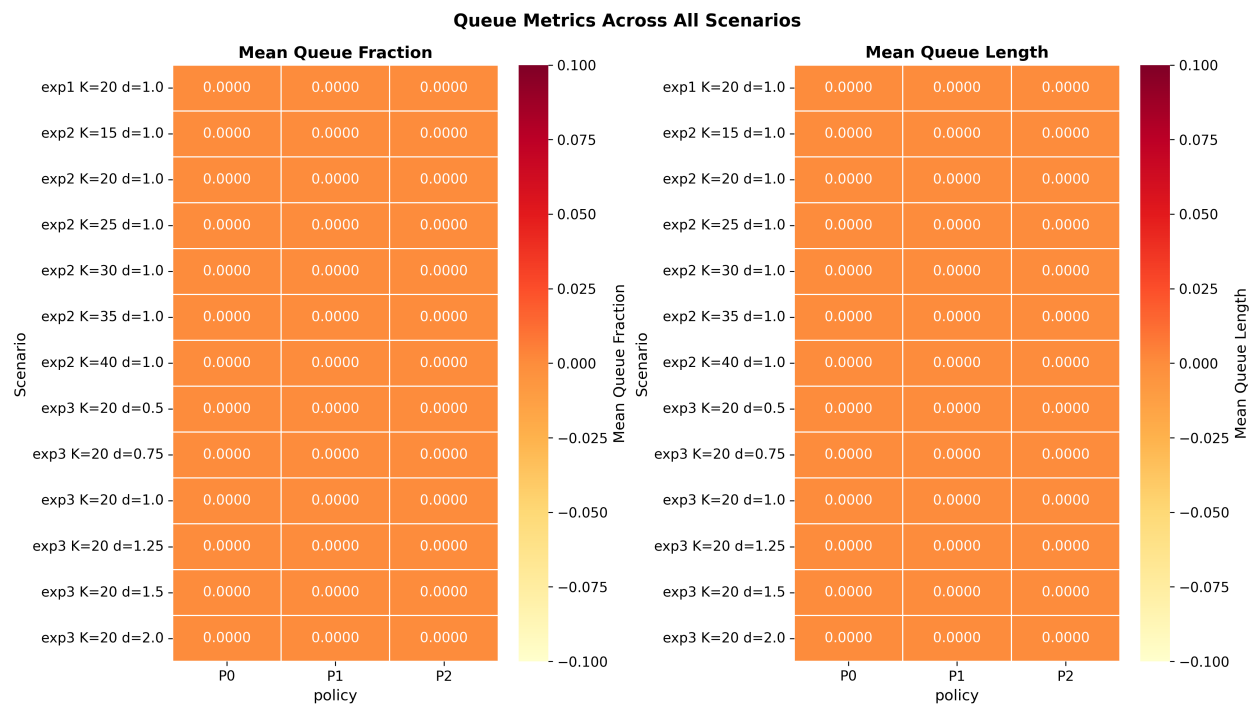
### Queue vs Fleet Size



## Queue vs Demand



## Queue Metrics Heatmap



## Conclusions

- Zero queueing across all experiments:** The EMS system has sufficient capacity to prevent incident queueing
- Utilization is the key indicator:** Low utilization (~10-15%) explains the absence of queueing
- Policy comparison is about spatial efficiency:** Since queueing is zero for all policies, the P2 advantage comes entirely from better spatial allocation
- Queue metrics validate capacity assumptions:** The absence of queueing confirms that the tested fleet sizes are adequate for Manhattan's crash demand

## Files Generated

| File                                                        | Description                    |
|-------------------------------------------------------------|--------------------------------|
| results/baseline/tables/<br>queue_statistics.csv            | Comprehensive queue statistics |
| results/baseline/tables/queue_anova.csv                     | ANOVA test results             |
| results/analysis/figures/<br>queue_comparison_by_policy.png | Policy comparison box plots    |
| results/analysis/figures/<br>queue_vs_fleet_size.png        | Fleet size sensitivity         |
| results/analysis/figures/<br>queue_vs_demand.png            | Demand sensitivity             |
| results/analysis/figures/queue_heatmap.png                  | Cross-scenario heatmap         |